

Python Skill Test

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Q1

Which method replaces a substring with another substring in Python?

Answer

replace()

Q2

What will be the output?

```
x = 5 + 3 * 2
```

```
print(x)
```

Answer

11

Q3

What will be the output?

```
s = "Hello123"
```

```
print(s.isalpha())
```

Answer

False

Q4

Write a Python code snippet to check even or odd.

Answer

```
if num % 2 == 0:  
    print("Even")
```

```
else:  
    print("Odd")
```

Q5

What is the default step value in range(start, stop, step)?

Answer

1

CODE_1:

Problem Statement

David is testing a new smart home control panel that can execute specific commands for lights and fans. The system processes a limited number of commands and must output the corresponding action exactly as given, or an error message if the command is not recognized.

The rules are:

- If the command is "LightsOn", print "LightsOn".
- If the command is "LightsOff", print "LightsOff".
- If the command is "FanOn", print "FanOn".
- If the command is "FanOff", print "FanOff".
- If the command is "Status", print "Status".
- For any other command, print "Invalid Command".

Input format :

The first line of input contains an integer n, representing the number of commands to process.

The next n lines each contain a string, representing a command.

Output format :

For each command, print the corresponding output as per the above rules.

Refer to the sample output for formatting specifications.

Code constraints :

The given test cases fall under the following constraints:

$$1 \leq n \leq 2$$

Commands are case sensitive and must match exactly.

Sample test cases :

Input 1 :

1

FanOff

Output 1 :

FanOff

Input 2 :

2

FanOff

LIGHTSOFF

Output 2 :

FanOff

Invalid Command

ANSWER:

```
# You are using Python
```

```
n=int(input())
```

```
for i in range(n):
```

```
    s=input()
```

```
    if s=='FanOff':
```

```
        print(s)
```

```
    elif s=='FanOn':
```

```
    print(s)
elif s=='LightsOn':
    print(s)
elif s=='LightsOff':
    print(s)
elif s=='Status':
    print(s)
else:
    print("Invalid Command")
```

CODE_2:

Problem Statement

In a quality control process at a manufacturing plant, each product has a unique identification number. The quality assurance team wants to analyze these numbers by summing the digits that appear in alternative digits odd positions (from right to left) of the product's ID. This helps them identify patterns in defect rates based on these summed values.

Given a product's identification number, write a program to calculate the sum of the digits that are in alternative digits when read from right to left.

Example

Input

123456

Output

12

Explanation

The digits at odd positions are 6, 4, and 2, which sum up to 12.

Input format :

The input consists of a single integer n , representing the product's identification number

Output format :

The output displays an integer representing the sum of the alternating digits starting from the first digit from right to left.

Refer to the sample output for formatting specifications.

Code constraints :

$$1 \leq n \leq 10^6$$

Sample test cases :**Input 1 :**

123456

Output 1 :

12

Input 2 :

4448

Output 2 :

12

```
def get_digits(n):  
    digits = []  
    while n > 0:  
        digits.append(n % 10) # take digits from right  
        n //= 10  
    return digits
```

```
def sum_alternate_digits(digits):  
    total = 0  
    for i in range(0, len(digits), 2): # 0,2,4...  
        total += digits[i]  
    return total
```

CODE_3:

Problem Statement

At the Pro Gym, a member wants to calculate the total cost for a combination of workout sessions. There are three types of sessions:

- Warm-up Sessions – Each session costs 5 dollars
- Core Workouts – Each session costs 10 dollars
- Cool-down Exercises – Each session costs 3 dollars

The user will input the session types one by one.

Write a program to count the number of each type of session and calculate the total cost.

Apply the following conditions:

- If the total number of sessions exceeds 10, apply a 10% discount on the total cost
- If the number of core workouts exceeds 5, subtract 5 dollars from the total cost

If both conditions are satisfied, apply the 10% discount first, followed by the 5-dollar discount.

Input format :

The input consists of integers entered one per line:

- 1 represents a warm-up session
- 2 represents a core workout
- 3 represents a cool-down exercise

The input ends when -1 is entered.

Output format :

The first line of output displays "**Warm-up Sessions:** " followed by the number of warm-up sessions done.

The second line displays "**Core Workouts:** " followed by the number of core workouts done.

The third line displays "**Cool-down Exercises:** " followed by the number of cool-down exercises done.

The fourth line displays "**Total Cost:** " followed by the calculated total cost formatted to two decimal places.

Refer to the sample output for formatting specifications.

Code constraints :

The given test cases fall under the following constraints:

- The input consists of values 1, 2, 3, and terminates with -1.

Sample test cases :

Input 1 :

1

2

3

1

2

3

-1

Output 1 :

Warm-up Sessions: 2

Core Workouts: 2

Cool-down Exercises: 2

Total Cost: 36.00

Input 2 :

2

2

3

1

3

2

1

-1

Output 2 :

Warm-up Sessions: 2

Core Workouts: 3

Cool-down Exercises: 2

Total Cost: 46.00

Input 3 :

1

1

1

1

1

1

2

2

2

2

2

2

3

3

-1

Output 3 :

Warm-up Sessions: 6

Core Workouts: 6

Cool-down Exercises: 2

Total Cost: 81.40

ANSWER:

w=0

c=0

co=0

count=0

while True:

 n=int(input())

 count+=1

 if(n==1):

 w+=1

 elif(n==2):

 c+=1

 elif(n==3):

 co+=1

 elif(n==-1):

 break

print("Warm-up Sessions:",w)

print("Core Workouts:",c)

print("Cool-down Exercises:",co)

total= (w*5)+(c*10)+(co*3)

if(count>=10):

```
total=total*0.90
if(c>=5):
    total=total-5
print("Total Cost: %.2f"%total)
```